



Proposed CEA Link — December 2026

How this rota would work from December 2026, and how it compares with today's link

Prepared 29 September 2026 · every figure is calculated from the rota, not entered by hand

PROPOSAL

Familiar Nine

No duty over nine hours, and most shift times ones people already work

CODE **F9-24-K31s**

FINGERPRINT **598a1294**

SET AGAINST today's 20-week link

CREATED 28 September 2026

- ✓ **24** lines, one per person
- ✓ **35h 00m** a week Mon–Sat, on average — exactly the contract
- ✓ longest run **6** days (limit 13)
- ✓ **0** rests under 12h
- ✓ **0** avoidable tiring patterns (today 5)
- ✓ all **11** December staffing rules met

What this is. Familiar Nine is a proposal for the CEA link on the December 2026 timetable: a 24-week rotation for 24 people, where today's link has 20. It is for discussion, not a decision — it has not yet been through the roster office or a union rep.

DOES IT MEET THE DECEMBER STAFFING LEVELS?

11 of 11

rules met

today's link meets 4 · each rule on page 6

CAN IT BE RUN WITHIN THE LIMITS?

Yes

inside every hard limit

shortest gap 14h20 (limit 12h) · most days in a row 6 (limit 13) · 35-hour contract exact

IS IT TIRING?

0

avoidable tiring patterns

today's link has 5 · cover weeks at their worst · page 7

WHAT DOES IT TAKE?

24 people

4 more than today's 20

4 cover weeks for leave and sickness · 10 on a Sunday as overtime (today 8)

ARE STAFF WHERE THE TRAINS ARE?

Closer

than today on weekdays, Saturday and Sunday

weekdays 27.7 (today 52.0) · Sat 15.0 (65.9) · Sun 26.4 (71.7) · 0 is a perfect match · page 5

6 in 24

full weekends off

one in 4 weeks — today one in 5 (4 in 20)

59 a year

finishes at 23:00 or later, each

today 39 · about one extra every 2 or 3 weeks

65 a year

starts at 06:20, the open, each

today 62 · about one extra every 17 or 18 weeks

Likely positives for staff

- A full weekend off one in 4 weeks (today one in 5)
- Never more than 6 days in a row (today 9)
- About 2 fewer days at work a year
- Every closing shift shorter — by about 30 to 45 minutes
- No shift longer than 9h00 (today 9h10)
- Shortest rest between shifts 14h20 (today 12h30)
- ...and 4 more — see page 2

What colleagues may be concerned about

- More late finishes — about one extra every 2 or 3 weeks each
- The average shift 6 minutes longer
- 8 new shift times to learn

Still to settle

- **Which link is used.** This is a proposal; colleagues' views come first.
- **Then who starts on which line.** The rota does not say who works which week.

What changes, in numbers

Today's 20-week link beside Familiar Nine. Green is better than today, amber is worse, unshaded is neither.

QUESTION	TODAY	FAMILIAR NINE	IN PLAIN WORDS
People on the link	20	24	4 more people, one per line
On duty each day — weekday · Saturday · Sunday	11·12 · 10 · 8	14 · 14 · 10	the December levels ask for 14 on a Saturday and 10 on a Sunday
On at the open — weekday · Sat · Sun	4 · 4 · 3	5 · 5 · 4	at least 4 every day — met
Still on duty at 22:00 — weekday · Sat · Sun	4 · 2 · 3	6 · 5 · 5	at least 5 every day — met
Through to the close — weekday · Sat · Sun	2 · 2 · 3	3 · 4 · 3	at least 3 every day — met
How closely staff follow the trains — weekday · Sat · Sun	52.0 · 65.9 · 71.7	27.7 · 15.0 · 26.4	lower is closer; 0 would be a perfect match — see page 5
Cover weeks — for leave and sickness	lines 1, 7, 12, 17	lines 1, 7, 13, 19	evenly spaced — one week in 6
Most days worked in a row	9	6	Chiltern's limit is 13 (written source to confirm)
Shortest gap between two shifts	12h30	14h20	the limit is 12 hours
Full weekends off	4 in 20	6 in 24	one in 4 weeks, against one in 5 today
Days at work a year, not counting Sundays	219	217	Sundays are overtime; a cover week counts as 4 days
Average shift · longest shift	8h14 · 9h10	8h20 · 9h00	the average is Monday to Saturday; the longest is on any day
Early shifts, shortest to longest	7h15–8h30	7h15–9h00	an early starts before 11:00 · any day of the week
Late shifts, shortest to longest	7h30–9h10	7h30–8h55	a late starts at 11:00 or after · any day of the week
Closing shift — weekday · Sat · Sun	8h40 · 9h10 · 8h55	8h10 · 8h40 · 8h10	the shift that locks up
How far the start time moves from one week to the next — on average	4h00	1h44	smaller is easier on the body clock
Finishing at 23:00 or later — each, a year	39	59	about one extra every 2 or 3 weeks
Starting at 06:20, the open — each, a year	62	65	about one extra every 17 or 18 weeks
Saturdays worked — each, a year	26	30	a rostered Saturday is paid at time and a quarter; cover weeks not counted
Sunday overtime to share — each, a year	21	22	Sundays are overtime, as today
Ticket office late shift on a Sunday — people	1	2	the December plan: both in the office 15:00–18:00, then one of them on the floor; today one 14:30–23:25 closer keeps it until 22:30
Different shift times	18	16 (8 worked today)	8 new to learn, on 16 of the 20 working weeks — on page 4, the times with no figure under today's link
Avoidable tiring patterns (ORR and rail-industry lists)	5	0	early starts and a weekly rotation come with every link
December staffing rules met	4 of 11	11 of 11	staffing levels confirmed verbally, 29 Sep 2026

The rest of this sheet — the workings

- 3 The rota** — all 24 weeks, day by day, with hours and days worked
- 4 A week, and every shift time** — beside the ones worked today
- 5 People on duty, hour by hour** — against the December trains
- 6 The limits and the eleven December rules** — each one, with today's link beside it
- 7 The fatigue checks** — the Office of Rail and Road's list, in full
- 8 How the figures are worked out**

THE ROTATION

The 24-line link

Sunday to Saturday per line; everyone moves down one line each week and line 24 goes back to line 1. Hours and days at the right, cover beneath.

LINE	SUN	MON	TUE	WED	THU	FRI	SAT	MON-SAT hours	ALL WEEK hours	DAYS WORKED Mon-Sat	TURNS Mon-Fri
1	SP	SP	SP	SP	SP	SP	SP	35h 00m	35h 00m	—	—
2	14:30 22:30	14:30 23:25	RD	RD	06:20 14:20	07:00 16:00	06:20 15:00	34h 35m	42h 35m	4	3
3	07:15 15:30	06:20 14:20	RD	RD	06:20 14:20	06:20 14:20	06:20 15:00	32h 40m	40h 55m	4	1
4	RD	RD	06:20 14:50	07:00 16:00	06:20 13:35	06:20 13:35	06:20 15:00	40h 40m	40h 40m	5	3
5	RD	RD	07:00 16:00	07:00 16:00	07:00 16:00	07:00 16:00	RD	36h 00m	36h 00m	4	1
6	RD	06:20 14:20	06:20 14:20	06:20 14:20	RD	RD	07:30 16:30	33h 00m	33h 00m	4	1
7	SP	SP	SP	SP	SP	SP	SP	35h 00m	35h 00m	—	—
8	14:30 22:30	14:00 22:30	RD	RD	14:00 22:30	14:00 22:30	RD	25h 30m	33h 30m	3	1
9	RD	07:00 16:00	06:20 13:35	06:20 14:50	06:20 14:50	06:20 14:20	RD	41h 15m	41h 15m	5	4
10	RD	06:20 13:35	07:00 16:00	06:20 14:20	07:00 16:00	RD	06:20 14:50	41h 45m	41h 45m	5	3
11	07:15 15:30	06:20 14:50	06:20 14:50	RD	RD	06:20 14:50	07:30 16:30	34h 30m	42h 45m	4	1
12	07:15 15:30	06:20 14:50	06:20 14:20	06:20 13:35	RD	RD	15:15 23:55	32h 25m	40h 40m	4	3
13	SP	SP	SP	SP	SP	SP	SP	35h 00m	35h 00m	—	—
14	07:15 15:30	07:00 16:00	RD	RD	15:45 23:55	15:45 23:55	15:15 23:55	34h 00m	42h 15m	4	2
15	RD	RD	14:00 22:30	14:00 22:30	14:00 22:30	14:00 22:30	14:30 22:00	41h 30m	41h 30m	5	1
16	15:15 23:25	RD	RD	15:45 23:55	15:45 23:55	15:45 23:55	RD	24h 30m	32h 40m	3	1
17	RD	15:45 23:55	15:45 23:55	15:45 23:55	15:45 23:55	RD	14:30 22:00	40h 10m	40h 10m	5	1
18	15:15 23:25	15:45 23:55	15:45 23:55	RD	RD	15:45 23:55	15:15 23:55	33h 10m	41h 20m	4	1
19	SP	SP	SP	SP	SP	SP	SP	35h 00m	35h 00m	—	—
20	09:00 18:00	RD	RD	06:20 14:50	06:20 14:50	06:20 14:50	06:20 14:50	34h 00m	43h 00m	4	1
21	RD	RD	14:30 23:25	14:30 23:25	14:30 23:25	14:30 23:25	RD	35h 40m	35h 40m	4	1
22	RD	13:30 21:00	13:30 21:00	13:30 21:00	13:30 21:00	13:30 21:00	RD	37h 30m	37h 30m	5	1
23	RD	14:00 22:30	14:00 22:30	14:00 22:30	RD	RD	15:15 23:55	34h 10m	34h 10m	4	1
24	15:15 23:25	15:45 23:55	15:45 23:55	15:45 23:55	RD	RD	14:00 22:30	33h 00m	41h 10m	4	1
On duty	10 SP 4	14 SP 4	14 SP 4	14 SP 4	14 SP 4	14 SP 4	14 SP 4	35h 00m	38h 26m	4.20	14 one-turn weeks
Minutes	4,950	6,970	6,970	6,970	6,970	6,970	7,150	42,000	46,950		

Early turns Late turns Cover (spare) week — four duties of seven, any turn Rest day

Mon-Sat hours average 35h 00m over 24 lines, a cover week counted as a contracted week · days worked average over the 20 working lines · Turns: how many different shift times the week works Monday to Friday; **bold** marks a one-turn week (see below) · Minutes: total duty minutes each day — Monday to Saturday adds up to the contract (35 hours for each of the 20 working weeks = 42,000 minutes)

The words on this page

Link — the whole rota: 24 weeks laid out as 24 lines. Everyone works line 1, then line 2, and so on round the wheel; the link needs 24 people.

Line — one week of the pattern, Sunday to Saturday. You move down a line each week, and line 24 goes back to line 1.

Turn — a shift time, such as 06:20-13:35. An **early** starts before 11:00; a **late** starts at 11:00 or later.

Rest day (RD) — a day off inside the contracted week. Two together make a proper break; one on its own does not count as a 48-hour break.

Cover (spare) week — a week with no fixed times. You work four duties of the seven days, placed by the roster clerk to cover leave and sickness.

Sunday — not part of the contract. A Sunday duty is rest-day working, paid on top of the week, which is why the hours are shown Mon-Sat and all week.

The contract — 35 hours a week, averaged over the 24 lines; a cover week counts as a full contracted week.

One-turn week — one clock time Monday to Friday, with every day of the week an early or every day a late (shown bold in the Turns column); the easiest kind of week to live around.

The floor — staff out on the station (gates, concourse, platforms); the ticket office joins it only at the quiet ends (page 5).

Opener / closer — the first people on duty in the morning and the last at night.

Handover — the overlap when one shift takes over from another.

Match — how closely the number of people on the floor follows the number of trains through the day (page 5); 0 is a perfect match, lower is better.

A week, and the duty table

How a working week compares with today's, and every shift time beside the ones worked today.

The shape of a week, against today

MEASURE	TODAY (20)	PROPOSED (24)
Weeks on one shift time — all earlies or all lates	7 of 16	14 of 20
Weeks mixing early and late shifts	7	3
Rest-day breaks of two days or more (of all rest-day breaks)	13 of 17	22 of 24
Single rest days between duties (not a 48h break)	4	2
Days worked in a week (Sunday included; weeks × days)	1×3, 3×4, 11×5, 1×6	6×4, 14×5
Cover (spare) weeks (whole weeks)	lines 1, 7, 12, 17	lines 1, 7, 13, 19
Different shift times (8 not worked today)	18	16
Mon–Sat hours, average week (the contract)	35h 00m	35h 00m
Sunday duties (not contracted)	8 of 16 lines	10 of 20 lines

How many people are working

PEOPLE WORKING	TODAY	PROPOSED	WHY
Monday to Friday	11–12	14	24 people instead of 20, so more on each weekday
Saturday	10	14	meets the December 2026 figure of 14
Sunday	8	10	meets the December 2026 figure of 10
Ticket office Mon–Fri · Sat · Sun	4 · 4 · 3	4 · 4 · 4	ticket office only; today Sunday has one late, not two
Opening at 06:20 (07:15 Sunday) Mon–Fri · Sat · Sun	4 · 4 · 3	5 · 5 · 4	at least four to open, every day — met
Through to the close (23:55; 23:25 Sunday) Mon–Fri · Sat · Sun	2 · 2 · 3	3 · 4 · 3	at least three to close, every day — met
Still on duty at 22:00 Mon–Fri · Sat · Sun	4 · 2 · 3	6 · 5 · 5	at least five from 22:00, every day — met

The duty table, beside today's

SHIFT TIME	TODAY'S LINK			PROPOSED		
	MON-FRI	SAT	SUN	MON-FRI	SAT	SUN
06:20 - 13:35 7h 15m · Early	1			1		
06:20 - 13:45 7h 25m · Early	1			0 (–1)		
06:20 - 14:00 7h 40m · Early		1			0 (–1)	
06:20 - 14:20 8h 00m · Early	2			2		
06:20 - 14:50 8h 30m · Early		3		2 (+2)	2 (–1)	
06:20 - 15:00 8h 40m · Early					3 (+3)	
07:00 - 16:00 9h 00m · Early				2 (+2)		
07:15 - 15:30 8h 15m · Early						4 (+4)
07:15 - 15:45 8h 30m · Early			3		0 (–3)	
07:30 - 16:30 9h 00m · Early				2 (+2)		
08:00 - 16:30 8h 30m · Early	2			0 (–2)		
08:30 - 16:30 8h 00m · Early			1			0 (–1)
09:00 - 18:00 9h 00m · Early						1 (+1)
11:00 - 19:30 8h 30m · Late	1			0 (–1)		
12:00 - 20:00 8h 00m · Late		1			0 (–1)	
13:00 - 21:00 8h 00m · Late			1			0 (–1)
13:30 - 21:00 7h 30m · Late		1		1 (+1)	0 (–1)	
13:30 - 22:00 8h 30m · Late	1			0 (–1)		
14:00 - 22:30 8h 30m · Late	2			2	1 (+1)	
14:30 - 22:00 7h 30m · Late		2		2		
14:30 - 22:30 8h 00m · Late						2 (+2)
14:30 - 23:25 8h 55m · Late			3	1 (+1)		0 (–3)
14:45 - 23:55 9h 10m · Late		2		0 (–2)		
15:15 - 23:25 8h 10m · Late						3 (+3)
15:15 - 23:55 8h 40m · Late	2			0 (–2)	4 (+4)	
15:45 - 23:55 8h 10m · Late				3 (+3)		

People on each shift time on a Saturday, a Sunday and — under Mon–Fri — the weekday with most of that time, so Mon–Fri can add up to more than work on one day. Green: more than today; amber: fewer; grey italics: a time the proposal drops.

Monday to Friday add up to 34,850 minutes of duty and Saturday 7,150 — together exactly 42,000 minutes — 35 hours for each of the 20 working weeks, the contract (a cover week counts as a full 35-hour week on top). Sunday adds 4,950, paid on top of the contracted week. 8 of the 16 shift times are new — nobody works them today.

People on duty, hour by hour

Today's link and the proposal, hour by hour, against the December 2026 timetable. A darker orange hour carries more of the day's trains.

People on duty in that hour — darker blue is more. Anyone on for any part of the hour counts, so a handover counts both people.

Read down each column: the most people should be under the darkest orange.

Trains — carriages arriving and leaving in that hour; darker is busier. A guide from the timetable, not passenger numbers.

Match, on the right — how closely the floor follows the trains over the whole day. 0 is a perfect match; lower is better.

Monday to Friday

staffed 06:20–23:55 · a row for each different weekday

	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	Match
Today, Mon · Wed — all on duty		4	4	6	6	6	7	7	7	7	7	7	5	5	5	4	4	4	2	
Today, Tue · Thu · Fri — all on duty		4	4	6	6	6	7	7	8	8	8	8	6	6	6	5	5	4	2	
of whom in the ticket office		1	1	2	2	2	2	2	2	4	2	2	2	2	1	1	1	1		
of whom on the floor Mon · Wed		3	3	4	4	4	5	5	5	3	5	5	3	3	4	3	3	3	2	61.1
of whom on the floor Tue · Thu · Fri		3	3	4	4	4	5	5	6	4	6	6	4	4	5	4	4	3	2	50.1
Proposed — all on duty		5	7	7	7	7	7	7	8	10	9	7	7	7	7	7	6	6	4	
of whom in the ticket office		1	1	2	2	2	2	2	2	4	2	2	2	2	1	1	1	1		
of whom on the floor		4	6	5	5	5	5	5	6	6	7	5	5	5	6	6	5	5	4	27.7
Trains — carriages in and out	5	69	107	127	122	95	85	75	84	79	87	105	140	138	107	90	73	75	66	

Saturday

staffed 06:20–23:55

	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	Match
Today — all on duty		4	4	4	4	4	4	5	6	9	6	6	6	6	6	5	4	2	2	
of whom in the ticket office		1	1	2	2	2	2	2	2	4	2	2	2	2	1	1	1			
of whom on the floor		3	3	2	2	2	2	3	4	5	4	4	4	4	5	4	3	2	2	65.9
Proposed — all on duty		5	7	7	7	7	7	7	7	10	9	9	7	7	7	7	7	5	4	
of whom in the ticket office		1	1	2	2	2	2	2	2	4	2	2	2	2	1	1	1			
of whom on the floor		4	6	5	5	5	5	5	5	6	7	7	5	5	6	6	6	5	4	15.0
Trains — carriages in and out	6	24	58	75	81	92	88	77	64	68	72	84	82	67	64	79	78	62	41	

Sunday

staffed 07:15–23:25

	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	Match
Today — all on duty			3	4	4	4	4	4	5	8	8	5	4	4	4	4	3	3	3	
of whom in the ticket office			1	1	2	2	2	2	2	3	3	1	1	1	1	1	1	1		
of whom on the floor			2	3	2	2	2	2	3	5	5	4	3	3	3	3	2	2	3	71.7
Proposed — all on duty			4	4	5	5	5	5	5	7	10	6	6	5	5	5	5	5	3	
of whom in the ticket office			1	1	2	2	2	2	2	2	4	2	2	1	1	1	1	1		
of whom on the floor			3	3	3	3	3	3	3	5	6	4	4	4	4	4	4	4	3	26.4
Trains — carriages in and out			15	45	65	71	74	71	61	55	64	69	78	73	75	52	71	65	50	

Good to know

- The ticket office.** The December plan puts two people on every ticket-office shift, and they are not floor cover — except at the quiet ends of the day. Monday to Saturday one of each pair helps on the floor: in the morning from the open until 08:00, and in the evening from 19:30 until their office shift ends. That is why the “of whom in the ticket office” row drops to 1 in those hours and the floor row gains one. Where it shows 4, the early and late pairs are both there at the changeover. The “of whom” rows split the office out of each side, so the floor rows compare like with like.
- Sunday.** One early is on the floor until 09:00. The two lates are on the floor from the start of their shift until 15:00, then both in the office — at least 30 minutes’ handover before the earlies leave — and one goes back to the floor from 18:00 to the end of the shift. Today is different: one person keeps the office on a Sunday afternoon and evening — one of the three 14:30–23:25 closers, in the office until it shuts at 22:30 and then on the floor — so between 15:00 and 18:00 today's office row reads 1 where the proposal's reads 2.
- Only the floor is matched to the trains.** The ticket office is staffed to its opening hours, not to the trains, so its rows carry no match figure. Page 1's weekday figure is worked out from the five weekdays combined, so it is not an average of the scores shown.
- Reading the match.** It measures **shape, not numbers** — each hour's share of the day's staff against its share of the day's trains — so an extra person in a quiet hour makes it worse even though nobody is worse off. Read it beside the numbers of people, never alone.
- Cover weeks.** They are not in these figures — a cover week has no fixed times — so real cover is a little higher.
- Sunday's last trains.** On Sundays five trains in the December 2026 timetable arrive or leave after the 23:25 finish (the last at 23:54); the bar under the Sunday 23:00 hour marks the time after the finish. No duty runs past 23:25 on a Sunday — that is agreed practice and stays so (settled 28 Sep 2026) — so those trains fall outside the staffed day of any link, today's included.

Hard limits and the December 2026 rules

Hard limits first — met or not. Then the December 2026 rules, each met or not. The fatigue factors are on page 7.

Hard limits — a design either meets these or cannot be run

- ✓ **No more than 13 days worked in a row** — longest possible run **6** days (today 9)
Longest possible run is 6 days, within Chiltern's limit of 13. A cover week is 4 duties in 7 days and the roster clerk places them, so this is the worst case the link allows. Worked as a block of four rather than split day-on-day-off, the answer is the same — **6** days — wherever the block falls, so this row does not depend on how a cover week is placed.
The written source of the 13-day limit is still to be confirmed.
- ✓ **At least 12 hours between duties** — **0** rests under 12h anywhere in the rotation, Saturday-into-Sunday and line-into-line included (today 0)
Tightest anywhere: **14h 20m — line 2 Fri 07:00-16:00 into Sat 06:20-15:00** (today's tightest: 12h 30m).
- ✓ **The contracted week, exactly** — **35h 00m** average Mon-Sat over the 24 lines, cover weeks counted as contracted weeks
700h of duty a week across 20 working lines. Sunday duties (82h 30m in all) are paid on top as rest-day working, as they are today. Individual weeks range 24h 30m–41h 45m; only the average is the contract.

The December 2026 rules — each one, with today's link beside it

RULE	TODAY'S LINK	THIS PROPOSAL	WHAT THE RULE ASKS
At least four on at the open, every day	✗ 4 · 4 · 3	✓ 5 · 5 · 4	from 06:20 (07:15 on a Sunday)
At least three through to the close, every day	✗ 2 · 2 · 3	✓ 3 · 4 · 3	until 23:55 (23:25 on a Sunday)
At least five still on duty at 22:00, every day	✗ 4 · 2 · 3	✓ 6 · 5 · 5	someone finishing at 22:00 does not count
Fourteen on a Saturday, ten on a Sunday	✗ 10 · 8	✓ 14 · 10	
Four cover weeks, evenly spread	✗ lines 1, 7, 12, 17	✓ lines 1, 7, 13, 19	evenly means every 6 lines, e.g. 1, 7, 13, 19
The ticket office: two early and two late, identical shifts, every day	✗ Mon-Fri yes · Sat yes · Sun no	✓ Mon-Fri yes · Sat yes · Sun yes	Mon-Fri 06:20–14:20 and 14:00–22:30; Sat 06:20–14:50 and 14:30–22:00; Sun two from 07:15 and two to 22:30
Every weekday closer starts at 15:45	✗ closers start at 15:15	✓ closers start at 15:45	
At least two on the floor at every moment	✓ fewest 2 · 2 · 2	✓ fewest 3 · 3 · 3	the ticket office counts only while its staff help on the floor — Mon-Sat one of each pair until 08:00 and from 19:30; on a Sunday one early until 09:00, both lates until 15:00 and one from 18:00; checked every five minutes
Handovers: 15 minutes to each closer, 20 in the ticket office (30 on a Sunday)	✓ floor met · ticket office 20 min	✓ floor met · ticket office 20 min	the Sunday office handover runs from 15:00, when the lates join it; on a Sunday each opener also stays until 15 minutes after the last closer arrives; where the design does not roster the office pairs, the plan's posts are assumed
Sunday duties between 8h and 9h	✓ 8h 00m–8h 55m	✓ 8h 00m–9h 00m	
No more shift times than today	✓ 18	✓ 16	today 18

Still to settle

- **Nothing to settle on the rules** — all 11 are met.
- **New shift times.** 8 of the 16 are times nobody works today; page 4 lists them.

ORR fatigue factors

From the ORR's good-practice guidance, *Fatigue Factors*, page 3 (December 2021). $\triangle$ present — the pattern is in this design and worth a look · $\checkmark$ clear — it is not · $\bullet$ standing — comes with the station's hours or with any rotating link, not with this design's choices · – means it does not apply here.

These are 25 roster patterns that tend to tire people — 21 from the Office of Rail and Road's good-practice list and 4 checks from the rail industry's fatigue guidance (MRSF) — long runs of earlies, short gaps between duties, start times that jump about. For each one this table asks whether the pattern is in the rotation, today and proposed, and how big it is. The list is guidance: a factor present is a question to discuss, never a pass or a fail, and a design showing nothing is not thereby approved.

CODE	FACTOR	TODAY'S LINK	PROPOSED
FF2	Early shift starting 05:00–07:00 Time of day	$\bullet$ 24	$\bullet$ 30
FF3	Very early shift starting before 05:00 Time of day	$\checkmark$ 0	$\checkmark$ 0
FF5	Day shift over 12h Duty length	$\checkmark$ 0	$\checkmark$ 0
FF7	Early shift over 10h Duty length	$\checkmark$ 0	$\checkmark$ 0
FF8b	Fewer than 2 days rest after a block of early starts Recovery time	$\triangle$ 3	$\checkmark$ 0
FF11	More than 13 consecutive shifts without a 48h break Recovery time	$\checkmark$ 12 $\checkmark$ 9 as rostered	$\checkmark$ 8 $\checkmark$ 8 as rostered
FF10	More than 4 consecutive 12h day shifts Cumulative	$\checkmark$ 0	$\checkmark$ 0
FF15	More than 4 consecutive early shifts in a rotating pattern Cumulative	$\triangle$ 6	$\checkmark$ 4
MRSF	More than 12 consecutive day shifts Cumulative	$\checkmark$ 9	$\checkmark$ 6
MRSF	More than 55 hours worked in any 7-day period Cumulative	$\triangle$ 60.8	$\checkmark$ 51.3
MRSF	More than 7 consecutive 8h shifts Cumulative	$\triangle$ 9	$\checkmark$ 6
FF17	Backward rotating pattern Circadian	$\checkmark$ 10 backward / 12 forward	$\checkmark$ 12 backward / 12 forward
FF18	Rotating pattern of about a week Circadian	$\bullet$ 20-line rotation · typically 4h 00m a week	$\bullet$ 24-line rotation · typically 1h 44m a week
FF19	Successive start times varying by more than 2 hours (definition to confirm) Circadian	$\triangle$ 6	$\checkmark$ 0
FF13	Less than 12h rest in any 24h Intervals between duties	$\checkmark$ 0	$\checkmark$ 0

Also on the list, and applying to neither link: time of day (FF1) · duty length (FF4) · cumulative (MRSF) · night shifts (FF6, FF8, FF9, FF12, FF14, FF16, FF20).

- Avoidable tiring patterns found: 0 (today 5).
- **Standing factors ($\bullet$) are recorded for information, not counted.** FF2 comes with the station's 06:20 open, and FF18 — the ORR's concern with a rotation that changes shift type about once a week (*Managing rail staff fatigue*, 7.68) — comes with every weekly link, which moves everyone one line a week by construction; its week-to-week step is shown as information.
- **How to read the table.** The figure beside each mark is the worst case found, for today's link as for the proposal, except where a row counts only the duties the rota fixes: the 55-hour row counts a cover week as no hours, and FF8b counts none of its days as early. "As rostered" under FF11 is the worst of the places a cover week's four duties could fall when worked together, which is usual. The MRSF rows are extra checks from the rail industry's guidance on managing staff fatigue, listed beside the ORR's own. FF2 and FF15 use the ORR's early — a start from 05:00 up to 06:59, not the 11:00 used elsewhere on this sheet; the station opens at 06:20, so every link has FF2. FF11 counts shifts between two-day breaks, so a single rest day does not reset it — that is why it can be longer than the most days worked in a row.
- **Cover weeks.** A cover week's four duties can fall in different ways, and on this design that moves the run of 8-hour shifts (5 if placed well). Where they fall is for the roster office. This page is an aid to discussion, not a fatigue risk assessment.

How the figures are worked out

In plain words — what each figure means and what it does not.

Where the figures come from

Every figure is worked out from the rota itself — the 24 weeks on page 3 — by the Marylebone Roster app. None is typed in by hand, and the same calculation is run on today's link.

Cover weeks

A cover week is marked on all seven days but works four of them, placed by the roster clerk. It counts as a full contracted week of hours and as four days at work. Where a figure depends on where those four fall, the worst case is shown. Cover-week shift times are not known in advance, so the yearly counts of early and late starts, Saturdays and Sundays, and the rest gaps between shifts, cover the fixed duties only; whatever is later given in a cover week must still be rostered within the normal limits.

Sundays

Sunday is not in the contract: Sunday duties are overtime, as today. So "days at work" and the 35-hour week are Monday to Saturday. A Sunday off costs no annual leave.

Each person, a year

Figures given "each, a year" are averages across the whole link: every week's duties, times 52, divided by the number of people; these counts include Sunday overtime duties. "Days at work a year" is Monday to Saturday only: the days a week times $365 \div 7$. Somebody's own year depends on which week they start on.

Following the trains

For each hour, the share of the day's floor staff on duty is set against the share of the day's train movements in the December 2026 timetable.

Hour by hour, the difference between the two shares is squared — so a big mismatch counts far more than several small ones — and the squares are added up and multiplied by 10,000. 0 would be a perfect match; lower is closer. The staff share is worked out from the minutes each person spends on the floor, with the ticket office's fixed posts taken out — finer than the whole-person counts on page 5, where a person counts in every hour they are on duty for any part of, so a handover inside an hour counts both people. It measures the shape of the day, not a staffing requirement or passenger numbers, and it compares shares, not headcounts: an extra person in a quiet hour makes the figure worse although nobody is worse off. The ticket office is left out, except for the help described under "The ticket office".

The rota is supplied beside this PDF as Familiar-Nine-F9-24-K31s-import.txt, ready to paste into the Links page (Import), which re-runs every check here from the pasted weeks. The code **598a1294** in every footer is worked out from the 168 days of the rota and nothing else, so a printout always matches the design it came from.

The ticket office

The December plan has two people on every ticket-office shift. They are not floor cover, except at the quiet ends: Monday to Saturday one of each pair helps on the floor until 08:00 and from 19:30. On a Sunday one early helps until 09:00, and the two lates are on the floor until 15:00, then both in the office, one going back to the floor from 18:00. Today one person, not two, keeps the office on a Sunday afternoon and evening.

Rest, runs and weekends

The shortest gap is one gap — the tightest anywhere, Saturday into Sunday and the last week into the first included; a rota with many gaps just over 12 hours reads the same as one with none. The most days in a row counts every day with a duty, Sunday overtime included, and is the worst case, with a cover week's four duties placed as badly as they can be. A full weekend off is a Saturday off followed by a Sunday off.

Familiar shift times

How many of the shift times somebody already works today. A rough guide to how much there is to learn, not to what staff will accept.

Fatigue

The Office of Rail and Road's good-practice list of roster patterns that tend to tire people, with four rail-industry checks. It is guidance, not a pass or fail, and this is not a fatigue risk assessment. Two patterns — early starts at a 06:20 station, and a weekly rotation — come with every link, today's included, so they are recorded on page 7 and not counted. In the ORR's checks an early start is one from 05:00 up to 06:59; a "block" of earlies is two or more in a row; the start-time rows count how many times a start moves by more than two hours (a count, not hours); and the week-to-week move is the average change in a week's mean start time, Sundays included and cover weeks left out.

What is confirmed

The December 2026 staffing levels, a 24-person link and the Sunday cover were confirmed verbally on 29 September 2026. The written source of Chiltern's 13-day limit is still to be confirmed.

How firm each figure is

Firm: the rest gaps, the contract, the days in a row and the staffing counts — worked out from the rota and checked again by the app. Confirmed verbally: the December staffing levels. Advisory: the fatigue patterns — guidance, never a pass or fail. A guide: how closely staff follow the trains, and how familiar the shift times are.

What no page can tell you

Whether staff accept the new shift times; where the roster clerk places a cover week's four duties; how leave and sickness cover land; whether the pattern holds on Wembley event days, since it follows the train timetable rather than passenger numbers; and who starts on which week. Those are for the people who work it and the managers who roster it.